



**Universität
Zürich**^{UZH}

Department of Informatics
Dynamic and Distributed Information Systems Group

Normative Approaches to Diversity-Optimized Recommendations of Political News

Lucien Heitz, Ph.D. Defense, July 11, 2025

Advisor: Prof. Abraham Bernstein, Ph.D.

Reviewers: Prof. Dr. Juliane A. Lischka, Prof. Robin Burke, Ph.D.

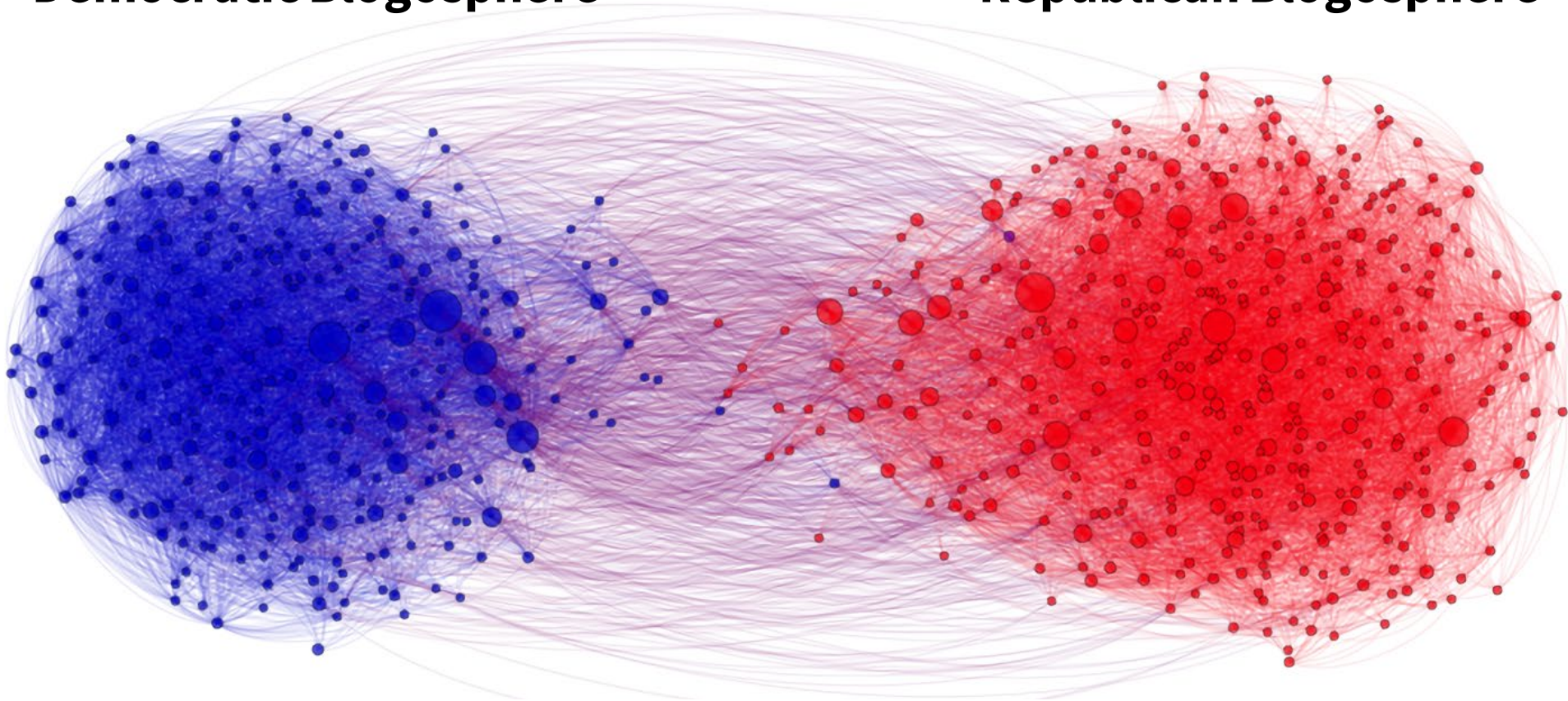
Chair: Prof. Dr. Michael Böhlen



Political Online News

Democratic Blogosphere

Republican Blogosphere



Visualizing Polarization in Political Blogs: <http://allthingsgraphed.com/2014/10/09/visualizing-political-polarization>



Research Goal

**Create diverse news recommendations
to counteract political polarization.**

WP1: Develop Diversity Algorithms

WP2: Build News App Infrastructure

WP3: Conduct User Experiments



RQ1: How to operationalize diversity for news?

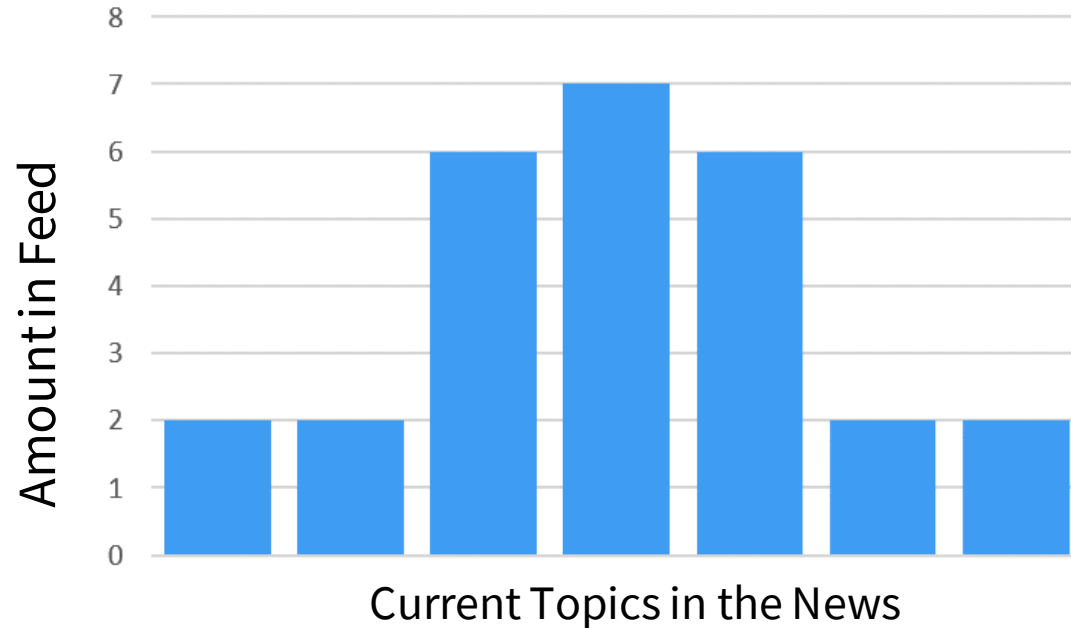
Based on:

Heitz et al., Informfully Recommenders – A Reproducibility Framework for Diversity-aware Intra-session Recommendations. Accepted at RecSys '25, 2025.



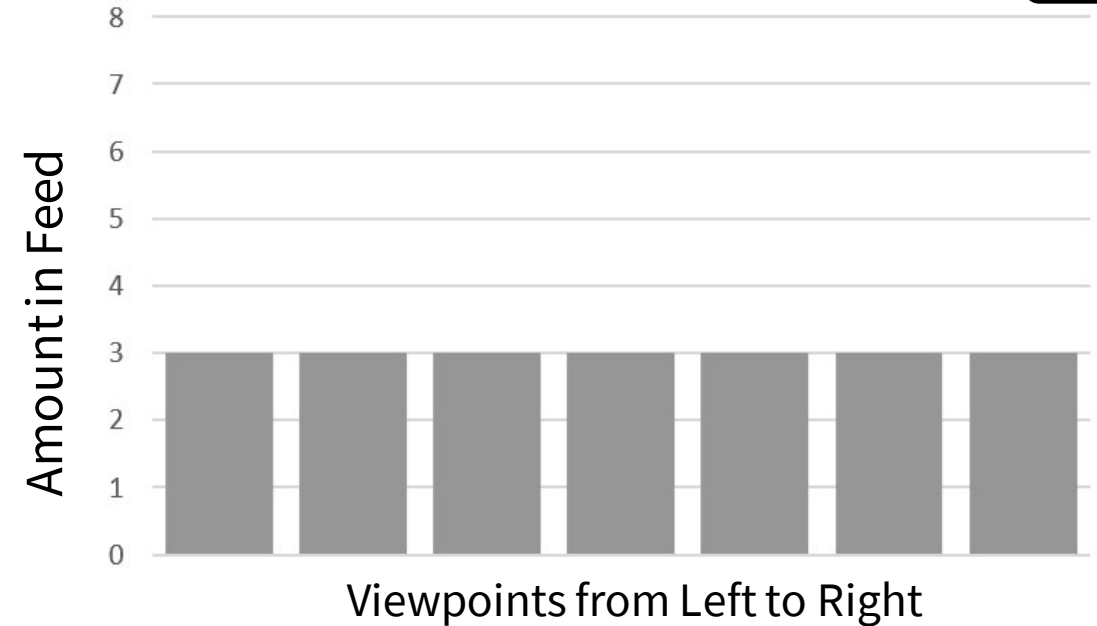
News Algorithms

Participatory Diversity



Goal: Discuss common issues.

Deliberative Diversity

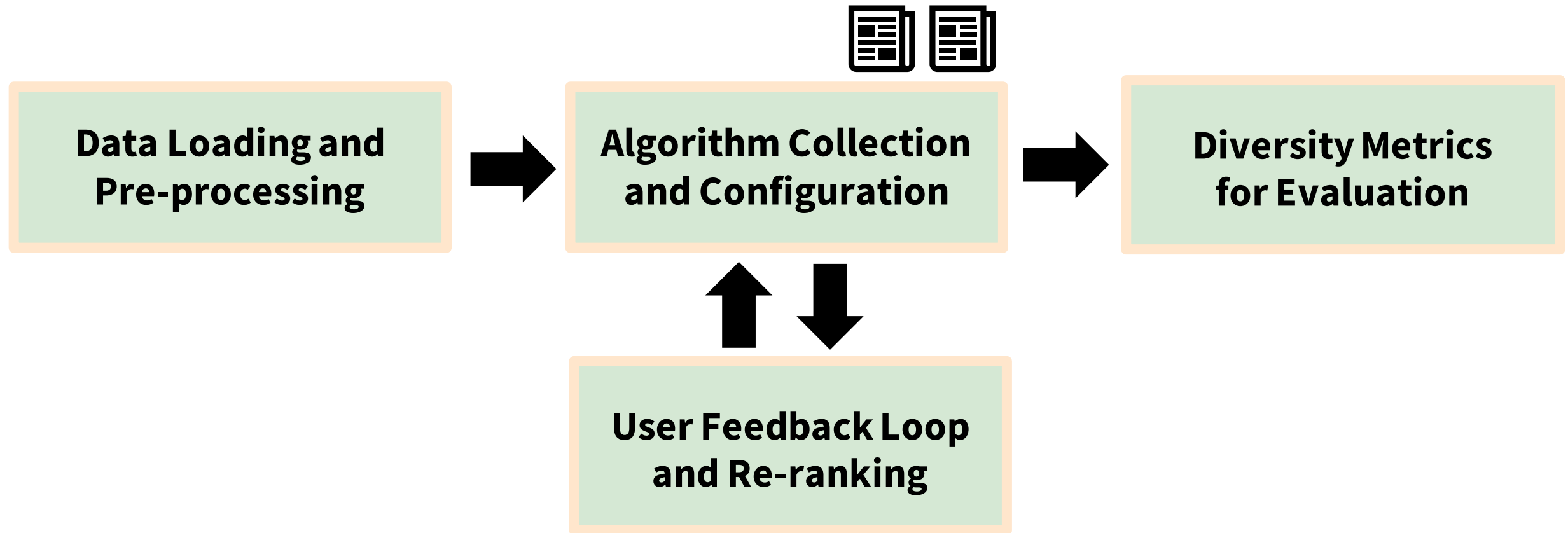


Goal: Give a voice to everyone.

Model goals based on Helberger, N.: On the Democratic Role of News Recommenders.
Algorithms, Automation, and News. Routledge, 2021.



Diversity Framework





Performance Evaluation

MIND	Fragmentation	Category Gini	Party Gini	Category ILD	Party ILD
LSTUR	0.503	0.855	0.905	0.618	0.247
NPA	<u>0.443</u>	0.792	0.854	0.751	0.323
NRMS	0.712	0.756	0.896	0.764	0.246
Part. Div. (PLD)	0.560	<u>0.658</u>	<u>0.795</u>	<u>0.857</u>	<u>0.403</u>
Delib. Div. (EPD)	0.446	0.726	0.850	0.823	0.377

EB-NeRD	Fragmentation	Category Gini	Party Gini	Category ILD	Party ILD
LSTUR	0.610	0.832	0.874	0.740	0.343
NPA	0.619	0.826	0.877	0.753	0.337
NRMS	0.632	0.791	0.880	0.794	0.326
Part. Div. (PLD)	<u>0.418</u>	0.801	0.749	0.782	0.556
Delib. Div. (EPD)	0.486	<u>0.773</u>	<u>0.667</u>	<u>0.802</u>	<u>0.610</u>

Fragmentation used as a proxy for polarization. Lower value means larger overlap of read articles within the group. Category used as proxy for topics, party for viewpoints. Low Gini/high ILD is associated with more diverse news.



RQ2: How to test online news algorithms?

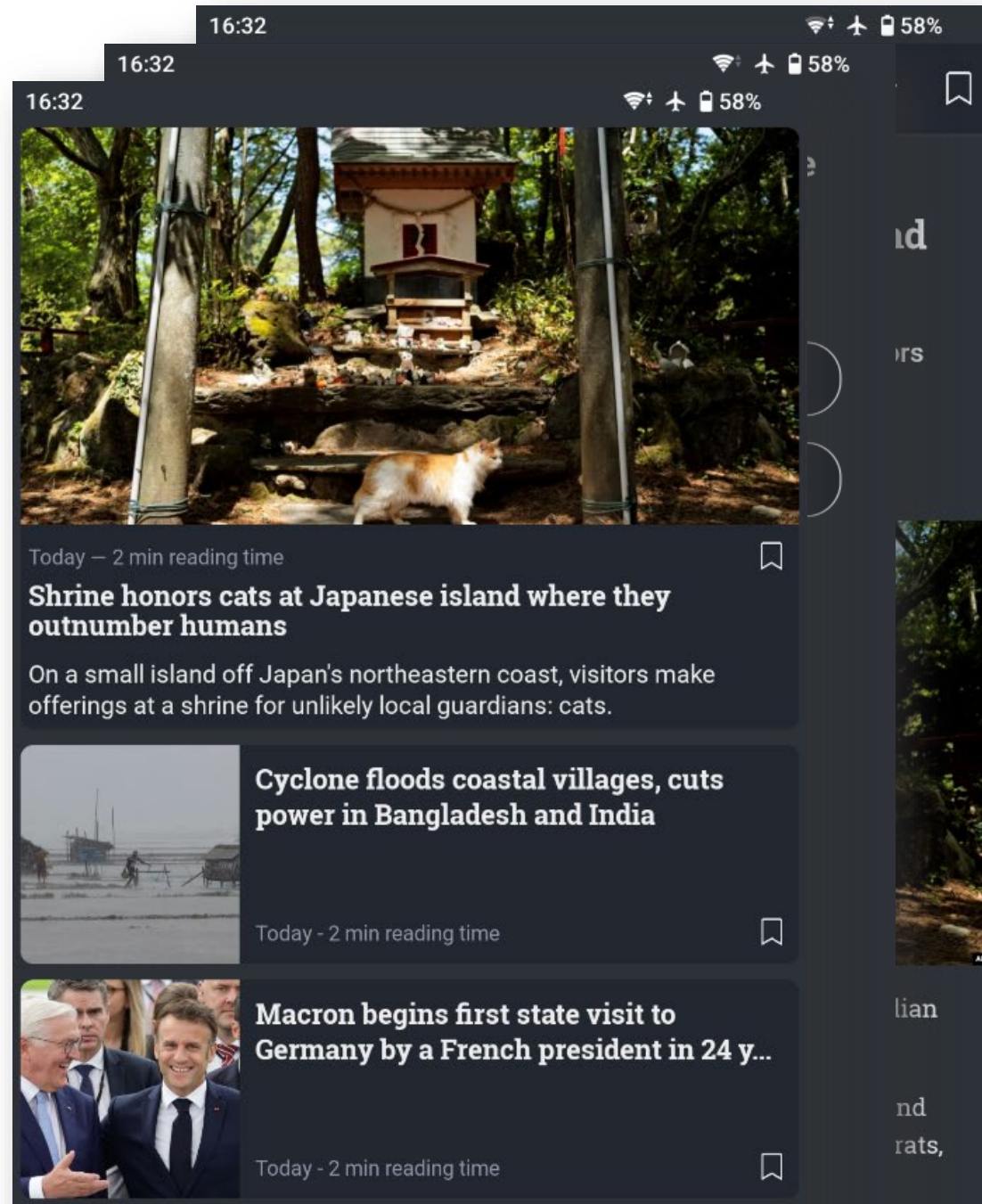
Based on:

Heitz et al., Informfully – Research Platform for
Reproducible User Studies. Published at RecSys '24, 2024.



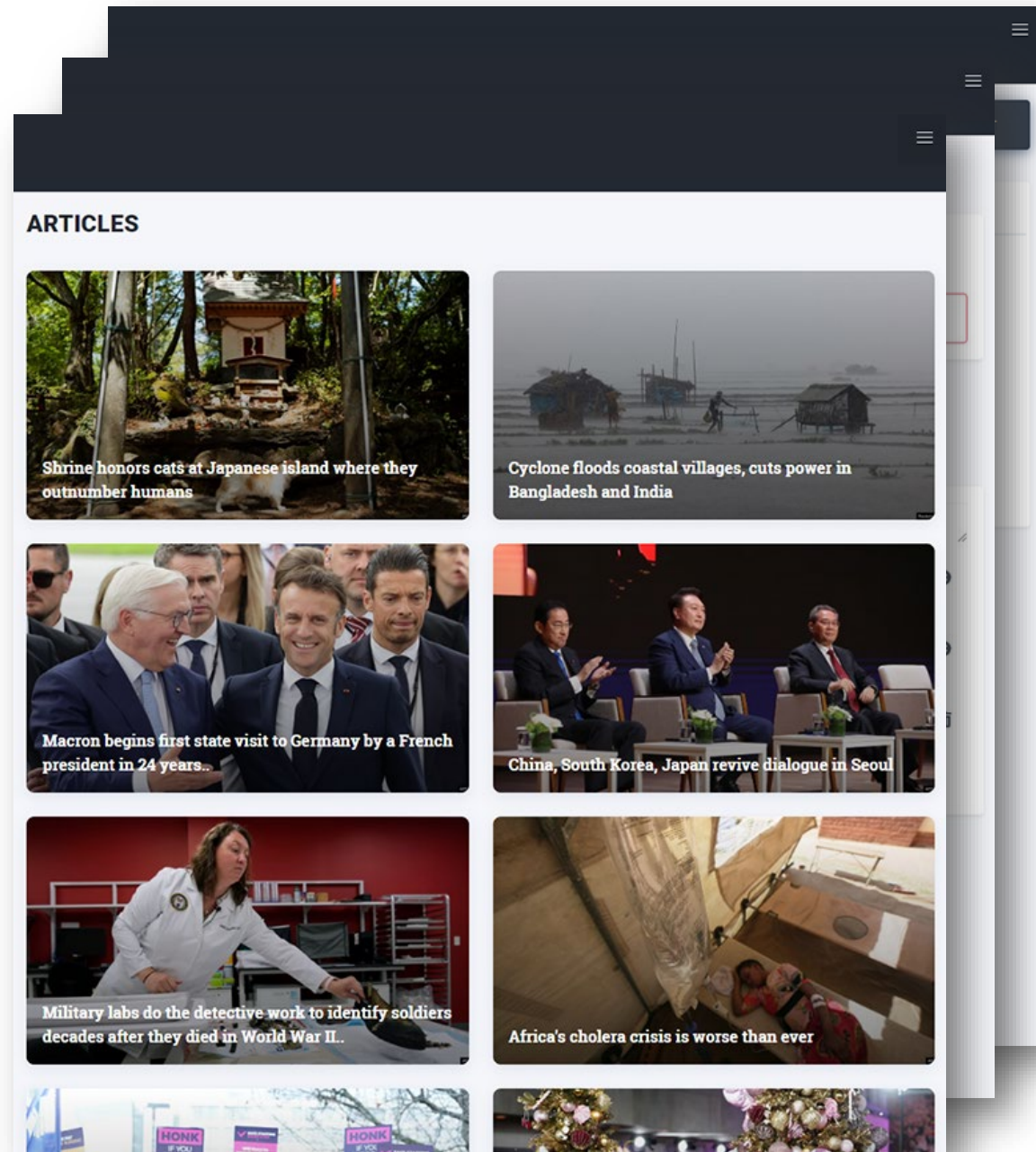
Mobile App

- Multimedia distribution platform
- In-app survey and tutorial section
- Android/iOS/Web



Admin Website

- Create user accounts
- Add and edit surveys
- Manage experiments





Feature Comparison

Tool	Platform	Media	Surveys	Open-source
PNS	Web	Text, Images	✗	✗
PEN	Web	Text, Images	✗	✗
Compendium	Android, iOS	Text, Images	✗	✗
3bij3	Web	Text	✗	✓
Blendle	Web	Text, Images	✗	✗
EasyStudy	Web	Text, Images	✗	✓
Informfully	Android, iOS, Web	Text, Images, Video	✓	✓



RQ3: What is the impact of diverse news?

Based on:

Heitz et al., Benefits of Diverse News Recommendations for Democracy: A User Study. Published in Digital Journalism 10(10), 2022.

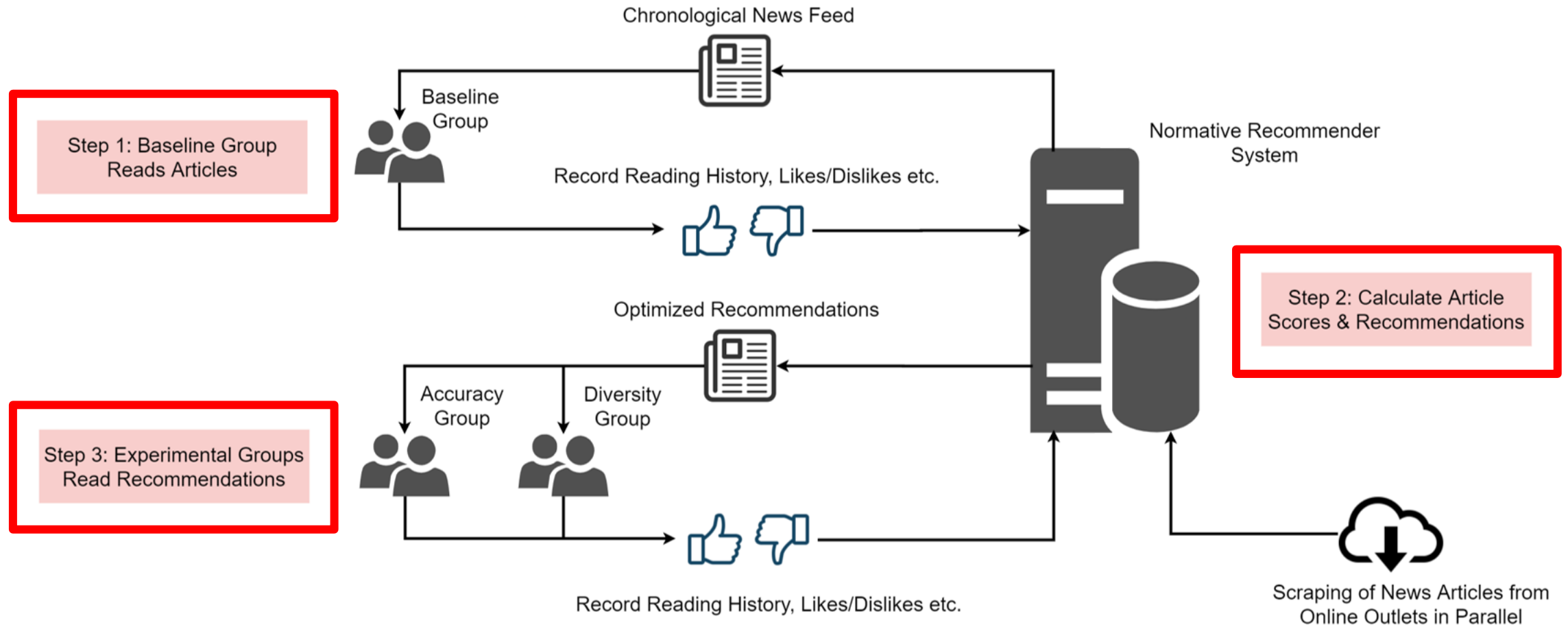
Heitz et al., Deliberative Diversity for News Recommendations: Operationalization and Experimental User Study. Published at RecSys '23, 2023.



User Studies

	Switzerland	Germany
Participants	151	143
User Groups	3	2
Duration (Days)	35	28
Outlets	6	6
News Items	7,287	2,191
User Interactions	55k	17k

Experiment Setting





Selected Experimental Findings

CH	Baseline	Accuracy	Part. Div.
Reading Time (min)	400	398	406
Political Knowledge	3.4	3.6	3.2
Opposing Views Pref.	3.7	3.1	4.8
Appreciation of Journalism	3.2	4.1	4.3

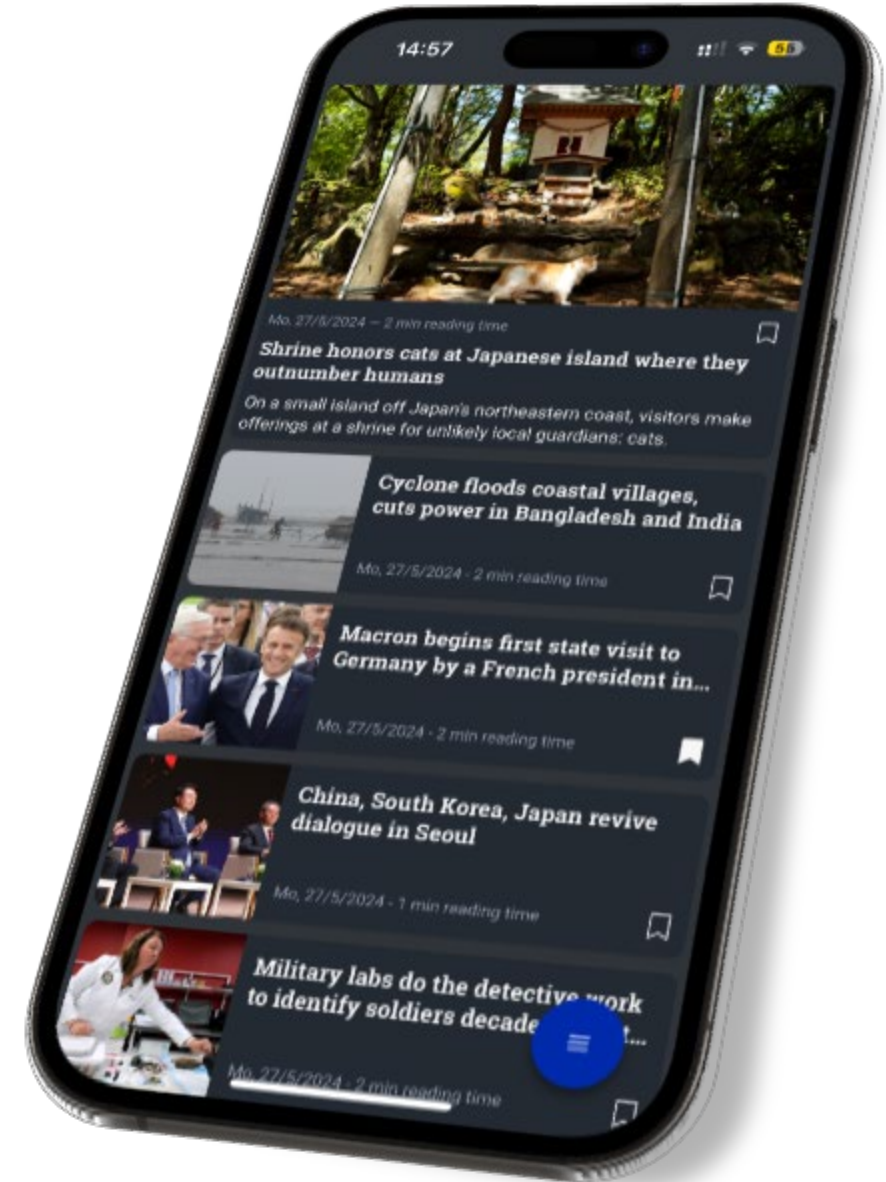
DE	Baseline	Delib. Div.
Reading Time (min)	216	302
Political Knowledge	82%	98%
Majority Support	12%	28%
Voting Participation	94%	89%

CH: Scores based on 7-point Liker scale self-assessment.
DE: Percentages based on yes/no questions and correct quiz answers (political knowledge).
Value significantly **higher** than the baseline. Value significantly **lower** than the baseline.



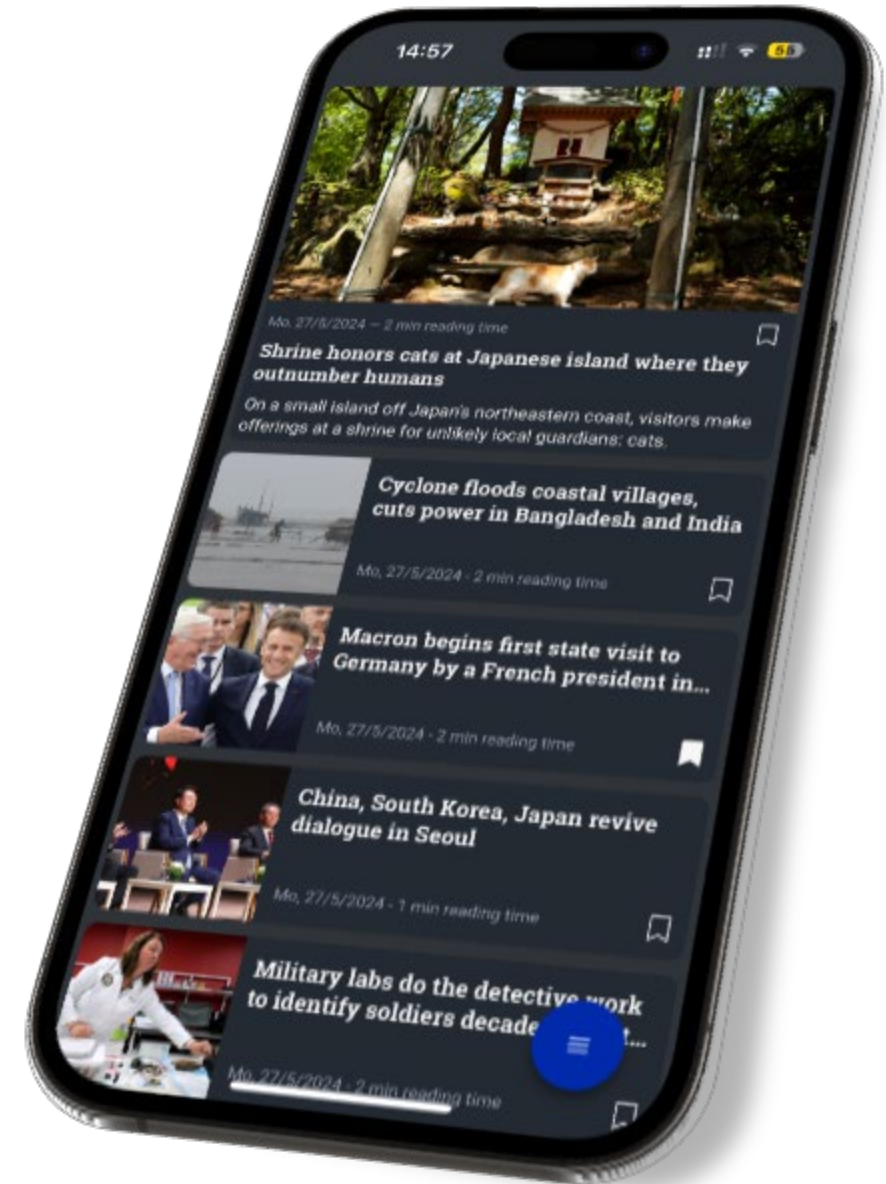
Summary

- Operationalization of normative diversity
- Development of a research infrastructure
- Empirical data on diversity impact



Limitations and Future Work

- Explore more diversity options
- Accommodate social features
- Generalization of the results





Thank You and Time for Q&A





Conclusion

Thank you!

Website: informfully.ch

GitHub: github.com/informfully

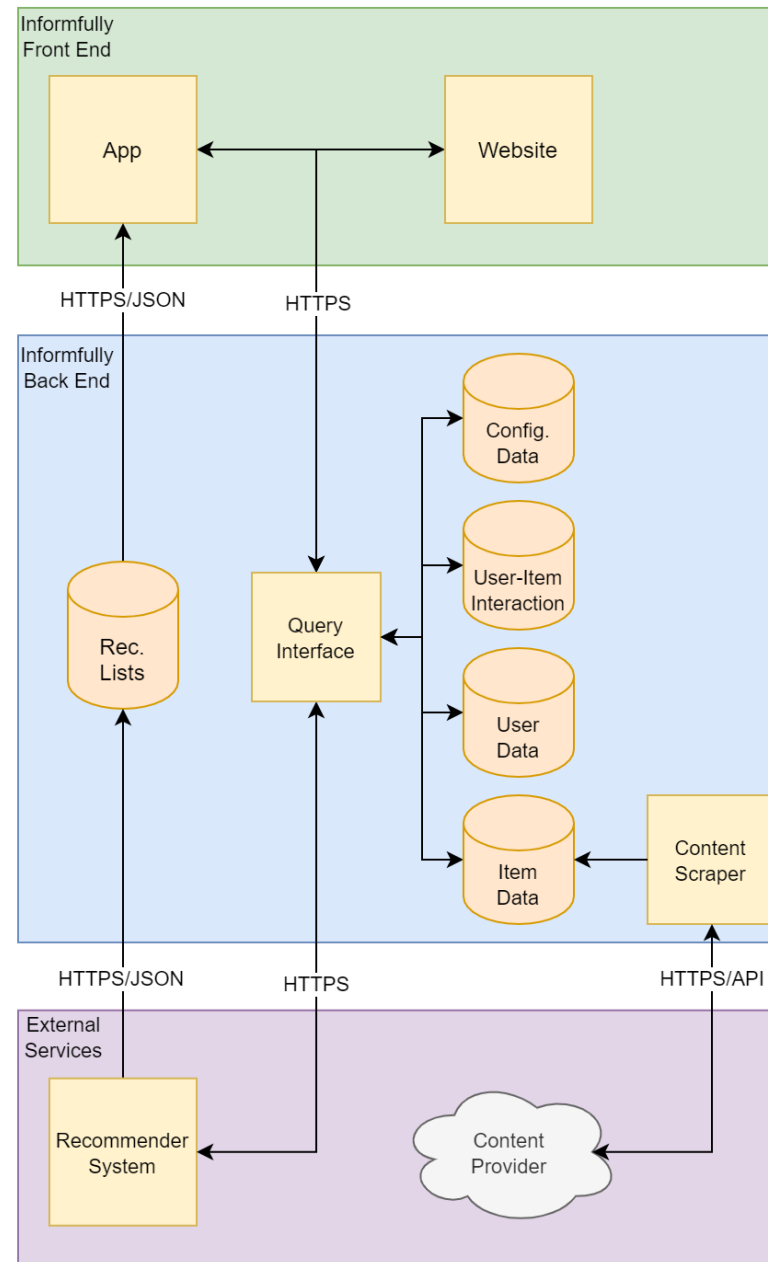
Email: heitz@ifi.uzh.ch

info@informfully.ch





Architecture



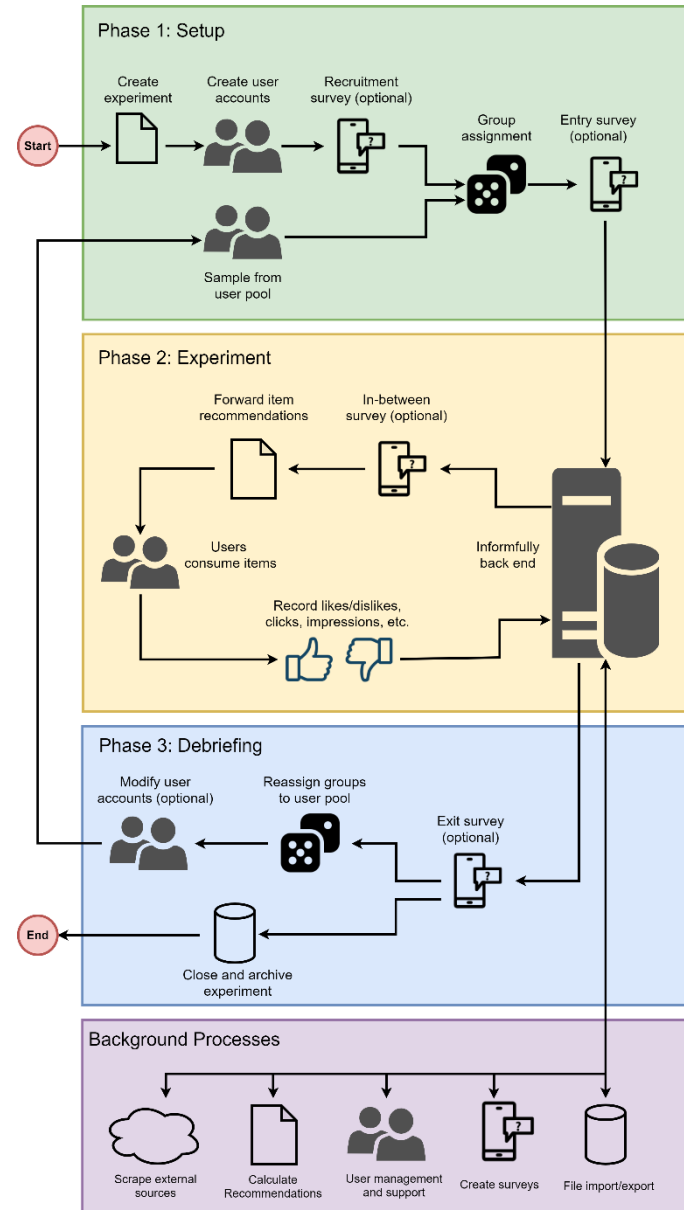


JREX

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1  {"recommendationLists": [  
2    {"ID": ObjectId("dbdwHPsadszYvgP"),  
3     "userID": "ksgsouZYPvBAGiQb",  
4     "itemID": "632013714366",  
5     "prediction": 0.117,  
6     "recommendationAlgorithm": "NMF",  
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8     "createdAt": 2024-05-21T12:19:40.229  
9   }  
10 ]}
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Experiments





The Role of News and Media

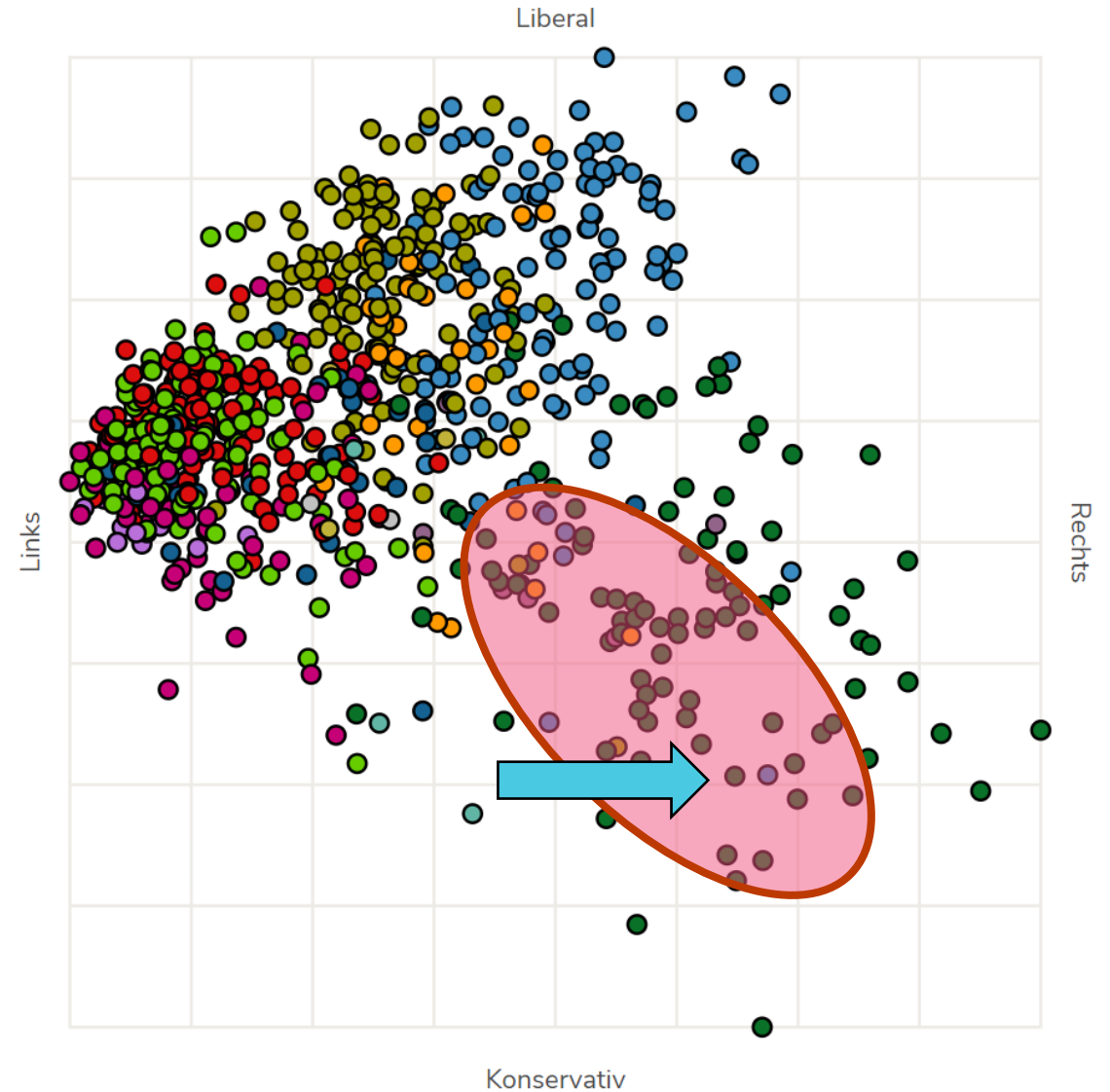
- Fourth Estate with control function
- Users, platforms, and society needs
- Create well-informed citizenry
- Interdisciplinary research field

Helberger, Natali. "On the democratic role of news recommenders."
Algorithms, automation, and news. Routledge, 2021. 14-33.



Optimizing for Political Diversity

- Recommendations of political news
- Map political space with Smartvote
- Place users and rate articles
- Define radius for algorithm



Smartmap from Smartvote survey:

https://www.smartvote.ch/de/group/38/election/22_vd_leg/smartmap

Diversity in Action with Informfully



Today - 2 min reading time

Shrine honors cats at Japanese island where they outnumber humans

On a small island off Japan's northeastern coast, visitors make offerings at a shrine for unlikely local guardians: cats.



Today - 2 min reading time

Cyclone floods coastal villages, cuts power in Bangladesh and India



Today - 2 min reading time

Macron begins first state visit to Germany by a French president in 24 years



Today - 1 min reading time

China, South Korea, Japan revive dialogue in Seoul



Today - 2 min reading time

Military labs do the detective work to identify soldiers decades after they die



Today - 2 min reading time

Africa's cholera crisis is worse than ever



Today, 03:12

Shrine honors cats at Japanese island where they outnumber humans

On a small island off Japan's northeastern coast, visitors make offerings at a shrine for unlikely local guardians: cats.

The 'Neko Jinja,' or Cat Shrine, mythologizes cats as guardian angels of Tashirojima, where cats outnumber humans.

Legend says the island used to be famous for sericulture and farmers would keep cats because they would chase away rats, protecting the silkworm cocoons from the rodents.

Fishermen on the island have also traditionally believed that cats bring good luck, including large hauls of fish.

Another legend says fishermen used to watch the cats' behavior for tips on the coming weather before heading to sea.

The islanders have long coexisted with the cats. One day, however, a fisherman accidentally injured a cat while working. Feeling sorry for the injury, the islanders built the shrine for cats.

Tashirojima is part of the city of Ishinomaki in Miyagi prefecture in the Tohoku region, which became well known after a tsunami devastated the area following a massive magnitude 9 earthquake on March 11, 2011.



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Your Feedback

I agree with the opinions expressed in the article.  0  0

I like the writing style.  0  0

I find the topic interesting.  0  0

More on the Subject



The high levels of toxic 'forever chemicals' in our fish could be harming..

Th, 12/14/2023 - 7 min reading time



World could be 7 years from gravest climate damage after 1.1% CO2 rise in 2...

Th, 12/14/2023 - 6 min reading time



Ban on cattle movement extended as more deadly bluetongue cases found in...

Th, 12/14/2023 - 3 min reading time

[Imprint](#)



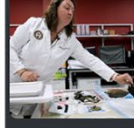
Today - 2 min reading time

Shrine honors cats at Japanese island where they outnumber humans



Today - 2 min reading time

Macron begins first state visit to Germany by a French president in 24 years



Today - 2 min reading time

Military labs do the detective work to identify soldiers decades after they die

What political party did you vote for in the last election?

Please select one answer

☐ Party A

☐ Party B



Open Challenges

- Balancing values and needs
- Lack of quality training data
- Design and user feedback
- Reproducible research



Informfully Platform

- Ease-of-use through seamless UI
- Versatility through interfaces
- Cost effective through templates
- Field-tested platform

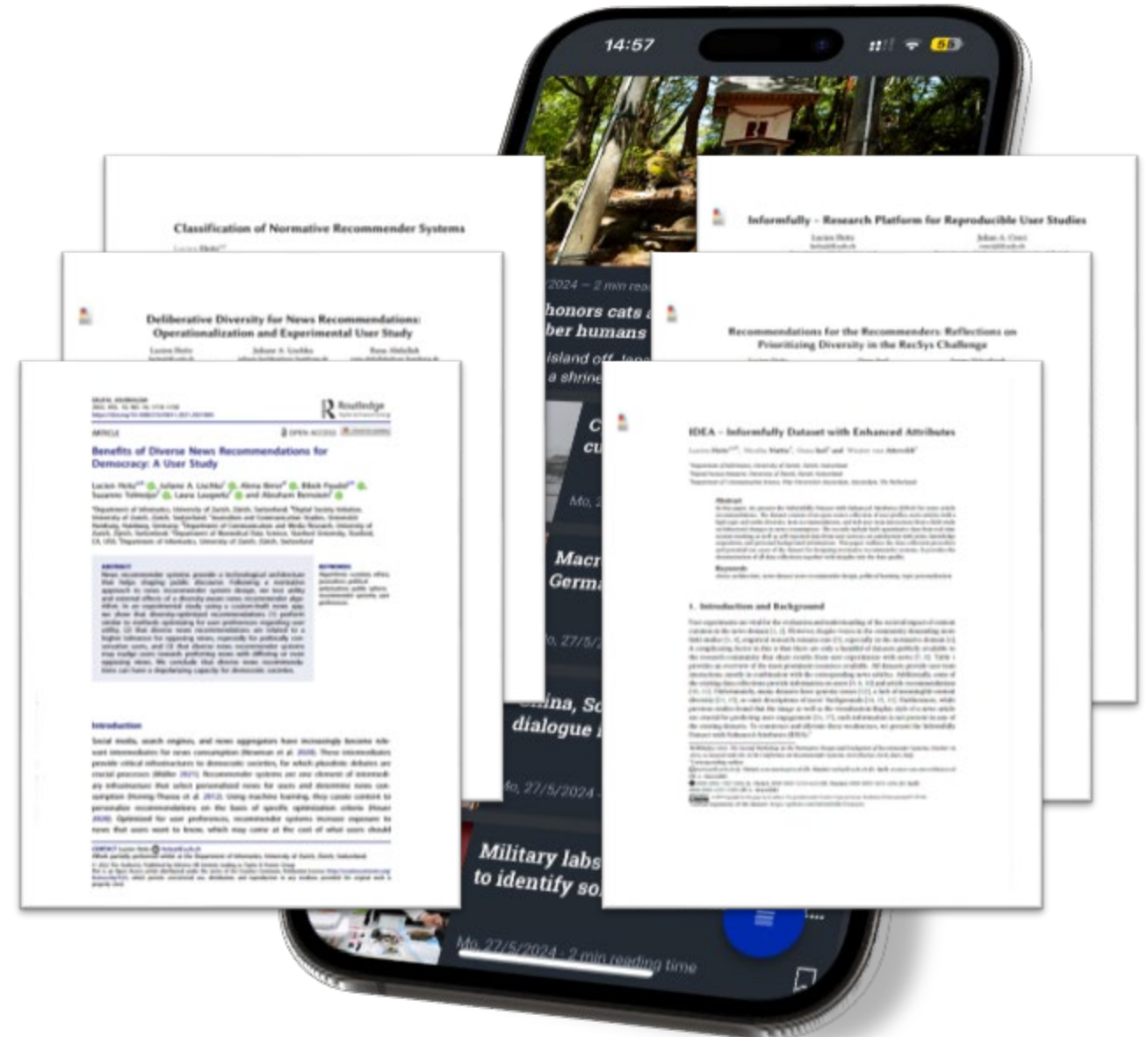


Table 3: Overview for *EB-NeRD* of the diversity scores for the top 20 news recommendations and AUC scores for predicting user impressions. The values closest to the perfect score are highlighted in green, second closest in blue, and third closest in red.

Model	Re-ranking	Activ.	Cat. Calib.	Comp. Calib.	Frag.	Alt. Voices	Repr.	Cat. Gini	Sent. Gini	Party Gini	Cat. ILD	Sent. ILD	Party ILD	AUC
LSTUR [3]	N/A	0.190	0.433	0.257	0.610	0.063	0.372	0.832	0.645	0.874	0.740	0.552	0.343	0.564
NPA [61]	N/A	0.202	0.468	0.266	0.619	0.061	0.365	0.826	0.605	0.877	0.753	0.580	0.337	0.554
NRMS [62]	N/A	0.204	0.509	0.285	0.632	0.060	0.362	0.791	0.544	0.880	0.794	0.624	0.326	0.549
LSTUR + NTD	G-KL	0.282	0.444	0.240	0.612	0.104	0.546	0.828	0.133	0.250	0.754	0.779	0.789	N/A
	PM-2	0.284	0.440	0.244	0.604	0.115	0.546	0.819	0.150	0.250	0.766	0.776	0.789	N/A
	MMR	0.295	0.433	0.235	0.613	0.118	0.575	0.823	0.226	0.270	0.759	0.762	0.807	N/A
	POS	0.365	0.480	0.259	0.667	0.110	0.731	0.844	0.759	0.853	0.699	0.430	0.344	N/A
	ATT	0.371	0.476	0.258	0.667	0.113	0.743	0.840	0.754	0.850	0.712	0.436	0.350	N/A
NPA + NTD	G-KL	0.292	0.469	0.237	0.589	0.108	0.545	0.810	0.133	0.250	0.775	0.779	0.789	N/A
	PM-2	0.302	0.471	0.239	0.576	0.122	0.547	0.806	0.150	0.250	0.783	0.776	0.789	N/A
	MMR	0.314	0.463	0.236	0.598	0.110	0.575	0.808	0.219	0.270	0.772	0.763	0.807	N/A
	POS	0.370	0.487	0.263	0.656	0.108	0.727	0.841	0.755	0.863	0.707	0.431	0.327	N/A
	ATT	0.377	0.484	0.259	0.655	0.111	0.742	0.841	0.742	0.860	0.708	0.448	0.331	N/A
NRMS + NTD	G-KL	0.307	0.482	0.239	0.608	0.091	0.544	0.798	0.133	0.250	0.790	0.779	0.789	N/A
	PM-2	0.316	0.483	0.244	0.598	0.099	0.544	0.796	0.150	0.250	0.794	0.776	0.789	N/A
	MMR	0.315	0.477	0.238	0.616	0.099	0.571	0.798	0.204	0.270	0.790	0.767	0.807	N/A
	POS	0.366	0.480	0.263	0.657	0.110	0.731	0.841	0.765	0.851	0.701	0.419	0.344	N/A
	ATT	0.371	0.476	0.260	0.655	0.112	0.742	0.839	0.747	0.848	0.709	0.444	0.349	N/A
D-RDW	N/A	0.374	0.407	0.229	0.394	0.107	0.556	0.798	0.133	0.250	0.810	0.779	0.789	0.554
RP^3_β [15]	N/A	0.222	0.415	0.235	0.582	0.080	0.376	0.840	0.755	0.856	0.743	0.439	0.392	0.565
RWE-D [42]	N/A	0.256	0.435	0.222	0.443	0.100	0.372	0.857	0.802	0.842	0.735	0.377	0.433	0.554
PLD [25]	N/A	0.152	0.459	0.268	0.418	0.038	0.432	0.801	0.687	0.749	0.782	0.534	0.556	N/A
EPD [24]	N/A	0.139	0.443	0.207	0.486	0.081	0.505	0.773	0.611	0.667	0.802	0.577	0.610	N/A
Random	N/A	0.180	0.461	0.256	0.705	0.054	0.366	0.756	0.634	0.873	0.842	0.564	0.346	0.500

Table 4: Overview for *MIND* of the diversity scores for the top 20 news recommendations and AUC scores for predicting user impressions. The values closest to the perfect score are highlighted in green, second closest in blue, and third closest in red.

Model	Re-ranking	Activ.	Cat. Calib.	Comp. Calib.	Frag.	Alt. Voices	Repr.	Cat. Gini	Sent. Gini	Party Gini	Cat. ILD	Sent. ILD	Party ILD	AUC
LSTUR [3]	N/A	0.266	0.620	0.313	0.503	0.051	0.296	0.855	0.593	0.905	0.618	0.585	0.247	0.593
NPA [61]	N/A	0.191	0.574	0.329	0.443	0.076	0.280	0.792	0.614	0.854	0.751	0.587	0.323	0.595
NRMS [62]	N/A	0.211	0.559	0.313	0.712	0.071	0.305	0.756	0.554	0.896	0.764	0.615	0.246	0.626
LSTUR + NTD	G-KL	0.250	0.619	0.311	0.558	0.097	0.408	0.786	0.133	0.250	0.769	0.779	0.789	N/A
	PM-2	0.227	0.607	0.317	0.574	0.086	0.418	0.809	0.133	0.250	0.734	0.779	0.789	N/A
	MMR	0.287	0.604	0.334	0.594	0.102	0.471	0.766	0.000	0.000	0.781	0.789	0.842	N/A
	POS	0.299	0.631	0.338	0.693	0.120	0.538	0.790	0.777	0.623	0.693	0.393	0.642	N/A
	ATT	0.289	0.633	0.340	0.688	0.134	0.544	0.786	0.746	0.618	0.712	0.438	0.652	N/A
NPA + NTD	G-KL	0.226	0.585	0.327	0.458	0.098	0.434	0.829	0.133	0.250	0.691	0.779	0.789	N/A
	PM-2	0.251	0.599	0.331	0.465	0.117	0.438	0.858	0.133	0.250	0.651	0.779	0.789	N/A
	MMR	0.320	0.581	0.329	0.475	0.137	0.517	0.837	0.000	0.000	0.691	0.789	0.842	N/A
	POS	0.329	0.636	0.372	0.549	0.173	0.568	0.900	0.745	0.640	0.436	0.431	0.637	N/A
	ATT	0.331	0.639	0.372	0.556	0.171	0.563	0.896	0.740	0.649	0.449	0.439	0.628	N/A
NRMS + NTD	G-KL	0.232	0.563	0.315	0.692	0.075	0.402	0.713	0.133	0.250	0.812	0.779	0.789	N/A
	PM-2	0.238	0.560	0.310	0.706	0.079	0.390	0.729	0.133	0.250	0.801	0.779	0.789	N/A
	MMR	0.284	0.574	0.322	0.692	0.088	0.468	0.716	0.000	0.000	0.809	0.789	0.842	N/A
	POS	0.346	0.605	0.352	0.713	0.117	0.561	0.766	0.729	0.623	0.740	0.435	0.639	N/A
	ATT	0.333	0.603	0.352	0.712	0.119	0.567	0.770	0.713	0.613	0.736	0.461	0.649	N/A
D-RDW	N/A	0.281	0.557	0.303	0.515	0.103	0.377	0.722	0.133	0.250	0.822	0.779	0.789	0.525
RP^3_β [15]	N/A	0.215	0.543	0.312	0.709	0.074	0.308	0.737	0.540	0.904	0.783	0.622	0.230	0.532
RWE-D [42]	N/A	0.225	0.583	0.304	0.398	0.102	0.298	0.724	0.364	0.830	0.805	0.715	0.352	0.512
PLD [25]	N/A	0.146	0.580	0.345	0.560	0.059	0.336	0.658	0.484	0.795	0.857	0.665	0.403	N/A
EPD [24]	N/A	0.274	0.614	0.318	0.446	0.100	0.399	0.726	0.533	0.850	0.823	0.626	0.377	N/A
Random	N/A	0.197	0.618	0.314	0.702	0.057	0.301	0.663	0.503	0.897	0.861	0.649	0.251	0.498

Table 5: Overview for *NeMig* of the diversity scores for the top 20 news recommendations and AUC scores for predicting user impressions. The values closest to the perfect score are highlighted in green, second closest in blue, and third closest in red.

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NPA [61]	N/A	0.206	0.601	0.420	0.787	0.073	0.568	0.905	0.980	0.725	0.577	0.019	0.556	0.491
NRMS [62]	N/A	0.220	0.629	0.428	0.730	0.055	0.556	0.864	0.963	0.808	0.707	0.037	0.468	0.552
LSTUR + NTD	G-KL	0.282	0.604	0.385	0.758	0.045	0.565	0.904	0.000	0.250	0.618	0.526	0.789	N/A
	PM-2	0.288	0.605	0.385	0.763	0.046	0.565	0.902	0.000	0.250	0.622	0.526	0.789	N/A
	MMR	0.192	0.617	0.400	0.826	0.046	0.535	0.876	0.497	0.431	0.675	0.396	0.723	N/A
	POS	0.535	0.602	0.371	0.651	0.076	0.614	0.898	0.927	0.667	0.619	0.048	0.552	N/A
	ATT	0.543	0.605	0.364	0.576	0.078	0.611	0.896	0.967	0.652	0.632	0.022	0.579	N/A
NPA + NTD	G-KL	0.275	0.598	0.385	0.686	0.042	0.571	0.910	0.000	0.250	0.572	0.526	0.789	N/A
	PM-2	0.285	0.600	0.386	0.693	0.045	0.571	0.906	0.000	0.250	0.582	0.526	0.789	N/A
	MMR	0.234	0.594	0.398	0.736	0.056	0.549	0.913	0.499	0.415	0.548	0.395	0.739	N/A
	POS	0.540	0.599	0.373	0.664	0.080	0.615	0.902	0.923	0.707	0.608	0.051	0.510	N/A
	ATT	0.538	0.603	0.368	0.588	0.090	0.616	0.899	0.967	0.701	0.628	0.022	0.534	N/A
NRMS + NTD	G-KL	0.281	0.624	0.382	0.667	0.040	0.576	0.877	0.000	0.250	0.680	0.526	0.789	N/A
	PM-2	0.289	0.622	0.382	0.664	0.037	0.575	0.879	0.000	0.250	0.675	0.526	0.789	N/A
	MMR	0.219	0.626	0.411	0.703	0.070	0.552	0.870	0.500	0.452	0.685	0.395	0.705	N/A
	POS	0.540	0.611	0.367	0.609	0.086	0.620	0.877	0.935	0.716	0.671	0.043	0.510	N/A
	ATT	0.546	0.618	0.363	0.521	0.097	0.625	0.870	0.979	0.739	0.693	0.014	0.500	N/A
D-RDW	N/A	0.289	0.598	0.382	0.737	0.042	0.554	0.887	0.000	0.250	0.653	0.526	0.789	0.550
RP^3_β [15]	N/A	0.185	0.612	0.408	0.880	0.049	0.545	0.856	0.921	0.761	0.717	0.076	0.508	0.448
RWE-D [42]	N/A	0.186	0.633	0.416	0.877	0.055	0.554	0.850	0.943	0.801	0.762	0.055	0.445	0.451
PLD [25]	N/A	0.127	0.625	0.417	0.641	0.027	0.526	0.858	0.926	0.792	0.716	0.074	0.466	N/A
EPD [24]	N/A	0.218	0.615	0.433	0.579	0.058	0.587	0.876	0.900	0.733	0.660	0.097	0.558	N/A
Random	N/A	0.185	0.628	0.412	0.879	0.047	0.541	0.843	0.929	0.772	0.738	0.068	0.489	0.498